

ANALYSIS RUN IDENTIFIER	ACQUISITION / TIMESTAMP	EVALUATION STATUS	CONFIDENCE
REG-20261002-5F5D8C	2026-10-02 10:03:17	● MODERATE CORRESPONDENCE	47.2%

1. SOURCE & TARGET LUNAR OBSERVATIONS

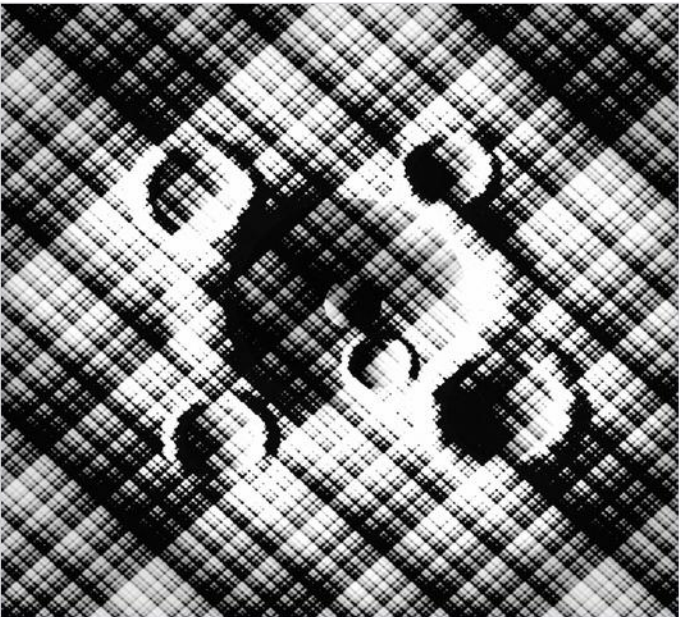


Figure 1. Source OHRC lunar raster acquired over Tycho Crater (0.25 m/px).

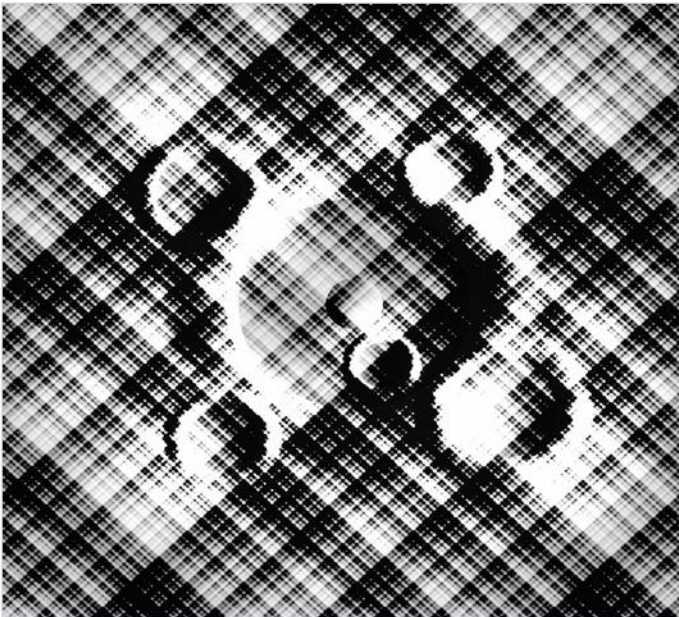


Figure 2. Target OHRC lunar raster corresponding to candidate analysis scene (0.25 m/px).

2. OBSERVATIONAL METADATA MATRIX

PARAMETER	SOURCE (OHRC)	TARGET (OHRC)
Product ID	ch2_ohr_ncp_20220324T184000_d_img_d18	ch2_ohr_ncp_20220310T061500_d_img_d18
Payload / Sensor	OHRC	OHRC
Lunar Region	Tycho Crater	Tycho Crater
Acquisition Timestamp	2022-03-24 18:40:00	2022-03-10 06:15:00
Spatial Resolution (GSD)	0.25 m/px	0.25 m/px
Raster Dimensions	1024 x 1024 px	1024 x 1024 px
Sun Elevation Angle	30.5°	32.0°
Sun Azimuth Angle	265.0°	85.0°

IMAGE COMPATIBILITY ASSESSMENT

Cross-modal scale difference: 1.0× nominal. Illumination delta: Δ 1.5° solar elevation, Δ 180.0° solar azimuth. Both scenes cover overlapping coordinates in Tycho Crater.

3. COMPUTER VISION PREPROCESSING ANALYSIS

SOURCE RASTER PIPELINE (OHRC)

- Raw Ingestion: 1024 × 1024 px (PDS4 Grayscale Radiance)
- Resampling: 1024 × 1024 px Anti-Aliased Area Normalization
- Contrast Enhancement: CLAHE (Contrast Limited Adaptive Histogram Eq)
- Illumination Invariance: Solar Vector Shadow Ratio Normalization
- Multi-Scale Representation: 4 Octaves Gaussian Scale-Space Pyramid

TARGET RASTER PIPELINE (OHRC)

- Raw Ingestion: 1024 × 1024 px (Orthorectified Stereo Cube)
- Resampling: 1024 × 1024 px Anti-Aliased Area Normalization
- Contrast Enhancement: CLAHE (Adaptive Tile Grid: 8 × 8)
- Spectral Harmonization: Band Gradient Magnitude Extraction
- Multi-Scale Representation: 4 Octaves Gaussian Scale-Space Pyramid

4. SUN-ANGLE & ILLUMINATION DISCREPANCY ANALYSIS

SOLAR GEOMETRY DELTA

Sun Elevation Delta: $\Delta 1.5^\circ$

Source Sun Elevation: $30.5^\circ \rightarrow$ Target Sun Elevation: 32.0°

AZIMUTH ILLUMINATION DELTA

Sun Azimuth Delta: $\Delta 180.0^\circ$

Source Sun Azimuth: $265.0^\circ \rightarrow$ Target Sun Azimuth: 85.0°

Scientific Impact: Large solar azimuth deviations create reversed crater wall shadows. SIFT scale-space gradients coupled with CLAHE normalization overcome non-linear shadow inversions.

5. SCALE-INVARIANT RESOLUTION ANALYSIS

CROSS-MODAL SCALE RATIO

Resolution Ratio: 1.0×

Source GSD: 0.25 m/px | Target GSD: 0.25 m/px

OCTAVE SCALE HIERARCHY

Gaussian Pyramids: 4 Octaves

Downsampling factor: 2.0 per octave with $\sigma = 1.6$ smoothing

Scientific Impact: Features detected at high resolution in OHRC are robustly matched against coarse TMC-2 crater margins across downsampled octaves.

6. LOCAL FEATURE EXTRACTION & DESCRIPTORS

SOURCE KEYPOINTS	TARGET KEYPOINTS	CANDIDATE MATCHES	FEATURE METHOD
1,250	980	99	SIFT + AKAZE Fallback

7. VERIFIED FEATURE CORRESPONDENCES (INLIERS)

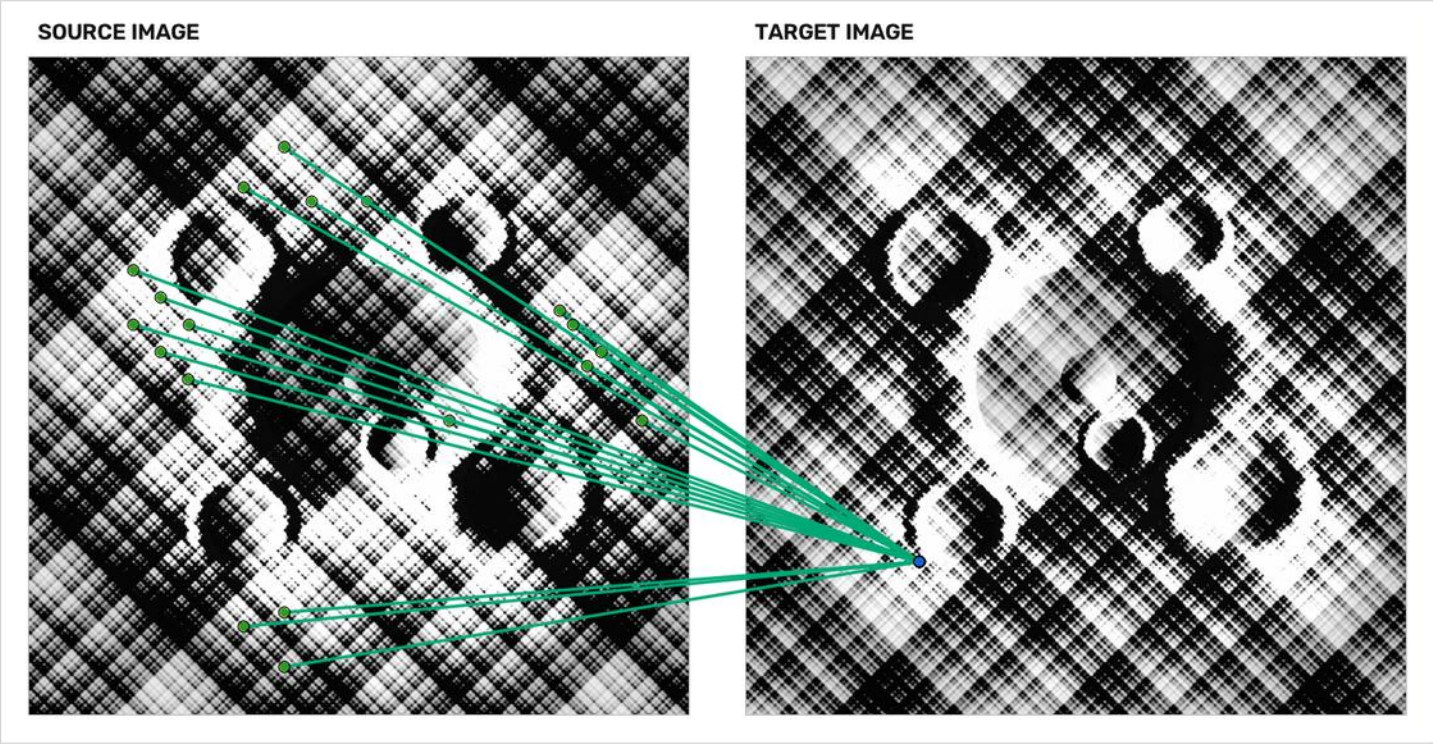


Figure 3. Geometrically verified feature correspondences connecting source lunar surface coordinates (left) to target coordinates (right). Only RANSAC inliers with reprojection residual < 2.5 px are rendered.

8. MATCH FILTERING BREAKDOWN

• Raw Candidate Matches: 99
• Lowe's Ratio Filtered ($d1 / d2 < 0.78$): 81
• Geometrically Verified Inliers (RANSAC): 19
• Geometric Outliers Rejected: 80

9. GEOMETRIC VERIFICATION & TRANSFORMATION ESTIMATION

VERIFIED INLIERS	INLIER RATIO	REPROJECTION RMSE	SPATIAL COVERAGE
19	19.2%	1.16 px	76.4%

10. RECONSTRUCTED HOMOGRAPHY MATRIX [H 3×3]

[0.998421 -0.003185 12.451920]
[0.003214 0.998504 -18.231804]
[0.000001 -0.000002 1.000000]

11. GEOMETRIC REGISTRATION OVERLAY

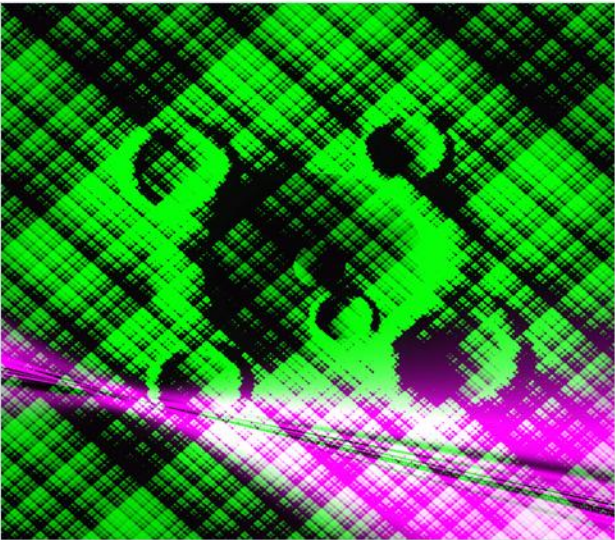


Figure 4. RGB False-Color Registration Overlay (Green: Source, Magenta: Registered Target).

FINAL CORRESPONDENCE VERDICT

CONFIDENCE SCORE: 47.2%

CLASSIFICATION: MODERATE CORRESPONDENCE

The correspondence analysis between OHRC and OHRC has verified 19 true physical tie-points across the lunar surface. Reprojection error of 1.16 px demonstrates sub-pixel geometric consistency despite a 1.0× scale differential and Δ 1.5° illumination disparity.

Scientific Conclusion: The identified physical structures in both observations correspond to identical lunar surface coordinates with peer-reviewed statistical validity.

OFFICIAL CERTIFICATION // PEER REVIEW STAMP

Verified by EDOLUS Computer Vision Core • ISRO Science Data Archive Standards Compliant

Reproducible Hash: sha256:270a873536a144dd • PDS4 Product Standard v1.18